

# Mathematical Preliminaries

Hyun Jae Stephen Chu\*  
University of Notre Dame

May 12th, 2026

## 1 Real Space Primer

We begin by studying several mathematical concepts on the **real line**  $\mathbb{R}$ . The real line is the simplest setting that contains several structures that will be important later:

- **Distance:** The quantity  $|x - y|$  measures how far apart two points are. This allows us to discuss neighborhoods, convergence, and continuity.
- **Order:** The relation  $x < y$  allows us to discuss upper and lower bounds, suprema, and infima.

The real line is also a convenient starting point because many of these concepts can be visualized. Once we understand the role played by distance on  $\mathbb{R}$ , we will extend the distance based concepts to more general spaces by replacing  $|x - y|$  with an abstract distance  $d(x, y)$ .

**Definition 1.** *The set  $\mathbb{R}$  denotes the set of all real numbers.*

Once we can measure distance, a natural question is “What does it mean for one point to be close to another point?” The word *close* has no fixed mathematical meaning. A distance of 0.1 may be considered small in one problem, but large in another. We therefore specify an allowed distance, denoted by  $\varepsilon > 0$ .

**Definition 2.** *The  $\varepsilon$ -ball around  $a \in \mathbb{R}$  for some  $\varepsilon > 0$  is the set of points  $x \in \mathbb{R}$  such that*

$$|a - x| < \varepsilon$$

*In other words,*

$$B_\varepsilon(a) = \{x: |a - x| < \varepsilon\}$$

Thus, an  $\varepsilon$ -ball is simply the interval consisting of all points whose distance from  $a$  is less than  $\varepsilon$ . The endpoints are excluded because the definition uses the strict inequality.

---

\*Comments and corrections are welcome and may be sent to [hchu@nd.edu](mailto:hchu@nd.edu).

On  $\mathbb{R}$ , an  $\varepsilon$ -ball is visualized as an interval. In  $\mathbb{R}^2$ , the analogous ball is the interior of a circle, and in more general metric spaces its shape depends on the chosen distance.

An  $\varepsilon$ -ball gives a somewhat local description since it tells us which points are close to one chosen point. We can now ask a corresponding global question about an entire set such that “Can all the points of the set be contained in some finite region, or does the set extend arbitrarily far along the real line?” This motivates the concept of boundedness.

**Definition 3.** A set  $A \in \mathbb{R}$  is called **bounded** if  $\exists M \in \mathbb{N}$  such that

$$|x| \leq M \forall x \in A$$

**Remark 1.** Note that the definition of boundedness of a set implies

$$|x| \leq M \iff -M \leq x \leq M$$

which is in fact

$$A \subseteq [-M, M]$$

Thus, we are drawing a finite window containing the entire set.

So far, we have considered sets of real numbers. A set records which values are present, but it does not record their order or how often they occur. In many mathematical problems, however, we have a value associated with each stage  $n \in \mathbb{N}$ . To express this ordering, we use a sequence.

**Definition 4.** A **real sequence** is a function from  $\mathbb{N}$  to a non-empty set  $X \subset \mathbb{R}$  with members denoted as  $x_n = x(n)$ .

Once a sequence has been specified, we can investigate the behavior of its terms as  $n$  becomes larger. Specifically, we might be interested in the question of “Do the terms of the sequence eventually remain arbitrarily close to one particular point?” The concept of an  $\varepsilon$ -ball allows us to state this particularly.

**Definition 5.** A real sequence is said to **converge** to a point  $x \in \mathbb{R}$  if  $\forall \varepsilon > 0 \exists N \in \mathbb{N}$  such that

$$|x_n - x| < \varepsilon \forall n \geq N$$

That point  $x$  is called a **limit** of the sequence.

**Theorem 1.** Each real sequence has at most one limit.

*Proof.* Suppose not and  $\{x_n\}$  has two limits  $x$  and  $x'$ . Then  $\exists N_1$  such that

$$|x_n - x| < \frac{\varepsilon}{2} \forall n \geq N_1$$

and  $\exists N_2$  such that

$$|x_n - x| < \frac{\varepsilon}{2} \forall n \geq N_2$$

Now take  $N = \max \{N_1, N_2\}$ . Then  $\forall n \geq N$ ,

$$\begin{aligned} |x - x'| &= |(x - x_n) + (x_n - x')| \\ &\leq |x - x_n| + |x_n - x'| \cdots \text{by triangular inequality} \\ &< \varepsilon \end{aligned}$$

This is a contradiction since  $\varepsilon$  is an arbitrary number that should be smaller than  $|x - x'|$ .  $\square$

Before introducing another type of sequence, it is useful to check that limits are compatible with the arithmetic operations on  $\mathbb{R}$ . If limits do not behave well under addition, multiplication, and other familiar operations, then knowing the limits of simpler sequences would not help us analyze more complicated ones.

**Theorem 2.** *Let  $\{x_n\}$  and  $\{y_n\}$  be real sequences. If  $x_n \rightarrow x$  and  $y_n \rightarrow y$ , then  $x_n + y_n \rightarrow x + y$ .*

*Proof.* Fix an arbitrary  $\varepsilon > 0$ . By triangular inequality,

$$|(x_n + y_n) - (x + y)| \leq |x_n - x| + |y_n - y|$$

Choose  $N_1$  such that  $|x_n - x| < \varepsilon/2$  for all  $n \geq N_1$  and  $N_2$  such that  $|y_n - y| < \varepsilon/2$  for all  $n \geq N_2$ . Let  $N = \max \{N_1, N_2\}$ . Then the RHS is less than  $\varepsilon$ , which implies that  $x_n + y_n \rightarrow x + y$ .  $\square$

To determine whether a sequence converges, the definition of convergence compares its terms with a proposed limit  $x$ . In some situations, however, we may not know the possible limit in advance. This motivates a different question where “do the terms of the sequence eventually become arbitrarily close to one another?” This question does not require us to identify a possible limit first and leads to the concept of a Cauchy sequence.

**Definition 6.** *A real sequence is **Cauchy** if  $\forall \varepsilon > 0 \exists N$  such that*

$$|x_n - x_m| < \varepsilon \text{ for all } n, m \geq N$$

The definition says that, after some stage  $N$ , every pair of remaining terms is within distance  $\varepsilon$  of each other.

**Remark 2.** *The following statement holds:*

*“Every convergent real sequence is Cauchy.”*

*This is because for a sequence  $x_n \rightarrow x$ , then sufficiently late terms are close to  $x$ , which implies that they must be close to each other.*

Then what about the converse?

**Theorem 3.** *In the real line, every Cauchy sequence is convergent. However, this is not true in a general metric space.*

The fact that every Cauchy sequence is not a convergent sequence in a general metric space later motivates the need for **completeness**.

Boundedness tells us whether an entire set can be contained within a finite interval. Since  $\mathbb{R}$  is also ordered, we can study if there is a number that is lying above or below every element of set  $A \subseteq \mathbb{R}$ .

**Definition 7.** Given a set  $A \subset \mathbb{R}$ , an **upper bound** of  $A$  is any number  $u$  such that  $a \leq u$  for all  $a \in A$ . Similarly, a **lower bound** of  $A$  is any number  $l$  such that  $l \leq a$  for all  $a \in A$ .

**Remark 3.** A set is bounded if and only if it is both bounded above and bounded below.

If the set contains the largest or smallest member, we call that member a maximum or minimum:

**Definition 8.** Let  $A \subseteq \mathbb{R}$ . We say that

- $a \in A$  is the **maximum** of  $A$  if  $a \in A$  and  $x \leq a \forall x \in A$ .
- $a \in A$  is the **minimum** of  $A$  if  $a \in A$  and  $a \leq x \forall x \in A$ .

However, a set does not necessarily need to contain a largest or smallest element. This motivates our need for supremum and infimum:

**Definition 9.** (Supremum and Infimum)

1. A number  $s \in \mathbb{R}$  is the **supremum** of  $A$  if
  - $s \leq u$  for all upper bounds  $u$ , i.e. is the least upper bound.
  - $\forall \varepsilon > 0, \exists x \in A$  such that  $x > s - \varepsilon$ .
2. A number  $i \in \mathbb{R}$  is the **infimum** of  $A$  if
  - $l \leq i$  for all lower bounds  $l$ , i.e. is the greatest lower bound.
  - $\forall \varepsilon > 0, \exists x \in A$  such that  $x < i + \varepsilon$ .

**Remark 4.** (Boundedness and Convergence)

- (**Monotone subsequence theorem**) Every real sequence has a monotone subsequence.
- (**Monotone convergence theorem**) Every bounded monotone sequence is convergent.
- (**Bolzano-Weierstrass**) Every bounded real sequence has a convergent subsequence.

**Definition 10.** Let  $f : A \rightarrow \mathbb{R}$  where  $A$  is nonempty. Then

$$\sup f \equiv \sup_{x \in A} f(x)$$

**Lemma 1.** Let  $f, g : A \rightarrow \mathbb{R}$ . Then

$$\sup (f + g) \leq \sup f + \sup g$$

*Proof.* By definition,  $\forall x \in A$ ,

$$f(x) \leq \sup f \text{ and } g(x) \leq \sup g$$

Then

$$\begin{aligned} f(x) + g(x) &\leq \sup f + \sup g \\ \Rightarrow \sup (f + g) &\leq \sup f + \sup g \end{aligned}$$

□

**Definition 11.** A function  $f : A \rightarrow \mathbb{R}$  is **bounded** if its range is a bounded set.

**Definition 12.** A function  $f : A \rightarrow \mathbb{R}$  is **continuous** if  $\forall x_n \in A$  such that  $x_n \rightarrow x \in A$ , we have  $f(x_n) \rightarrow f(x)$ .

**Theorem 4. (Intermediate Value Theorem)**

Let  $f : [a, b] \rightarrow \mathbb{R}$  where  $a < b$ . If  $f$  is continuous on  $[a, b]$  and  $f(a) < 0 < f(b)$ , then  $\exists c \in (a, b)$  such that  $f(c) = 0$ .

**Definition 13.** A function  $f$  is **concave** if  $\forall \lambda \in (0, 1)$ ,

$$f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y)$$

## 2 Exercises

**Exercise 1.** Prove that  $||a| - |b|| \leq |a - b|$  (Hint: The unremarkable identity  $a = a - b + b$  may be useful.)

**Exercise 2.** Show that, for an arbitrary function  $g : \mathbb{R} \rightarrow \mathbb{R}$ , it is always true that

$$g(A \cap B) \subseteq g(A) \cap g(B)$$

for all sets  $A, B \subseteq \mathbb{R}$ .

**Exercise 3.** Given sets  $A$  and  $B$ , define

$$A + B = \{a + b : a \in A, b \in B\}$$

Prove that if  $A$  and  $B$  are nonempty and bounded above then

$$\sup(A + B) = \sup A + \sup B$$

**Exercise 4.** If  $\sup A < \sup B$ , show that there exists an element  $b \in B$  that is an upper bound for  $A$ .

**Exercise 5.** Compute, without proofs, the suprema and infima (if they exist) of the following sets:

1.  $A \equiv \{\frac{m}{n} : m, n \in \mathbb{N} \text{ with } m < n\}$

2.  $B \equiv \{\frac{(-1)^m}{n} : m, n \in \mathbb{N}\}$

3.  $C \equiv \{\frac{n}{3n+1} : n \in \mathbb{N}\}$

**Exercise 6.** Let  $a < b$  be real numbers and consider the set  $T = \mathbb{Q} \cap [a, b]$ . Show  $\sup T = b$ .